How Dangerous Is Myanmar's Dam Infrastructure?

A Hobby Seismology Project After the 2025 M7.7 Mandalay Earthquake

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https://github.com/akzedevops/mmeqopendata

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Abstract

After the devastating March 2025 earthquake in Myanmar, I wanted to understand how many of the country's 254 dams are actually at risk. This write-up documents a hobby project that pulls earthquake data from a public API, scores every dam for seismic risk, and runs probabilistic hazard analysis—all with open-source Python tools. The corrected Abrahamson & Silva (2008) ground-motion model, validated against the actual Naypyidaw recording (predicted $0.51g$ vs. observed $0.57g$), shows that more than two-thirds of the dams sit in zones of significant seismic hazard. An OpenStreetMap-based exposure analysis finds over 2,100 schools and 900 hospitals in areas that experienced strong-to-severe shaking. Everything—code, data, interactive maps—is on GitHub.

Contents

1 Background

On March 28, 2025, a magnitude 7.7 earthquake hit central Myanmar along the Sagaing Fault. The rupture was enormous—475 km of surface break, supershear velocity, 5–7 m of slip. A seismic station in Naypyidaw recorded $0.57g$ of peak ground acceleration. Over 5,000 people died, and the government reported 586 dams damaged.

Myanmar has at least 254 documented dams (irrigation, hydropower, multi-purpose), many built before modern seismic codes existed. The Sagaing Fault runs right through the middle of the country where most of these dams are. I wanted to know: which dams are most at risk, and can we quantify it with open data?

1.1 What This Project Does

- Pulls 9,390 earthquake records (1970–2026) from the Myanmar Earthquake API
- Scores all 254 dams for seismic risk using a proper ground-motion model
- Runs probabilistic hazard curves, Coulomb stress transfer, fragility analysis
- Validates predictions against actual recorded ground motions
- Publishes everything as an open-source toolkit with interactive maps

1.2 Data Sources

All data is free and open:

Data	Source	License
Earthquakes	Myanmar Earthquake API (USGS/ISC)	Open
Dams (254)	Open Development Mekong (IFC/WLE)	CC BY-SA 4.0
Fault traces (395)	GEM Global Active Faults Database	CC BY-SA 4.0
Site V_{S30}	USGS ShakeMap V_{S30} grid (Wald & Allen 2007)	Public domain
Population grid	WorldPop 2020 (1 km)	CC BY 4.0
ShakeMap stations	USGS ShakeMap (us7000pn9s)	Public domain
Finite fault model	USGS (Goldberg et al. 2025)	Public domain
Surface rupture	USGS (Reitman et al. 2025)	Public domain
Buildings (34,224)	OpenStreetMap (Overpass API)	ODbL

2 The Earthquake Catalog

The catalog has 9,390 events. Most are small (mean M 3.2), but there are 6 events $\geq$M7.0 and 358 $\geq$M5.0. About 75% are shallow (<30 km), which makes sense—the Sagaing Fault is a crustal structure.

2.1 b-Value and Catalog Completeness

The Gutenberg-Richter law says $\log_{10} N = a - bM$. The tricky part is that the catalog is incomplete at small magnitudes—the 1970s network couldn't detect M2 events, but the modern network can. If you naively fit all the data, you get $b = 0.35$, which is way too low (global average is ~1.0). After Gardner-Knopoff (1974) declustering — magnitude-dependent space-time windows, e.g. ~86 km and ~967 days for the M7.7 mainshock — $M_c \approx 4.7$ and $b \approx 1.05$, typical for a tectonic region; these are the rates used for the probabilistic hazard below.

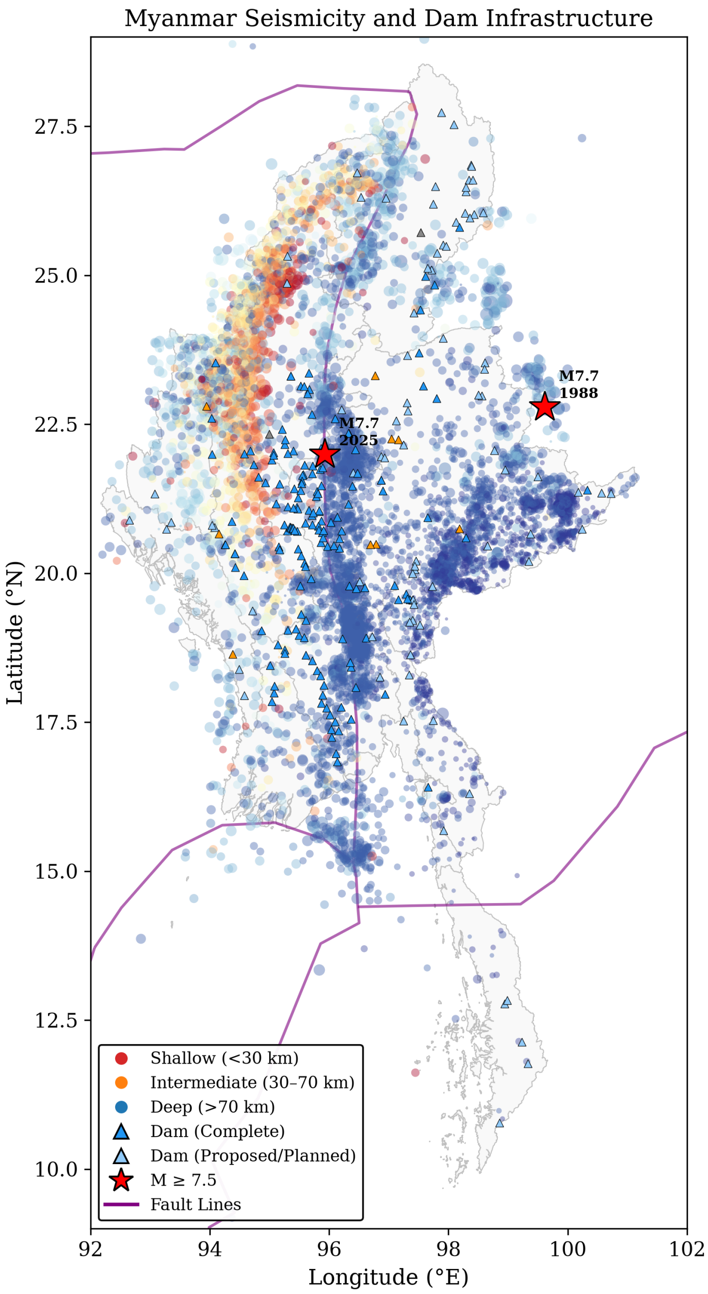

Figure 1: Earthquake distribution, dam locations, and fault lines. The clustering along the Sagaing Fault corridor is obvious.

A stability analysis (Figure 2b) shows the b-value only converges at $M_c \geq 4.0$. Using that cutoff: $b = 0.71 \pm 0.02$ with 2,473 events. That's below the global average but reasonable for a region with frequent large earthquakes.

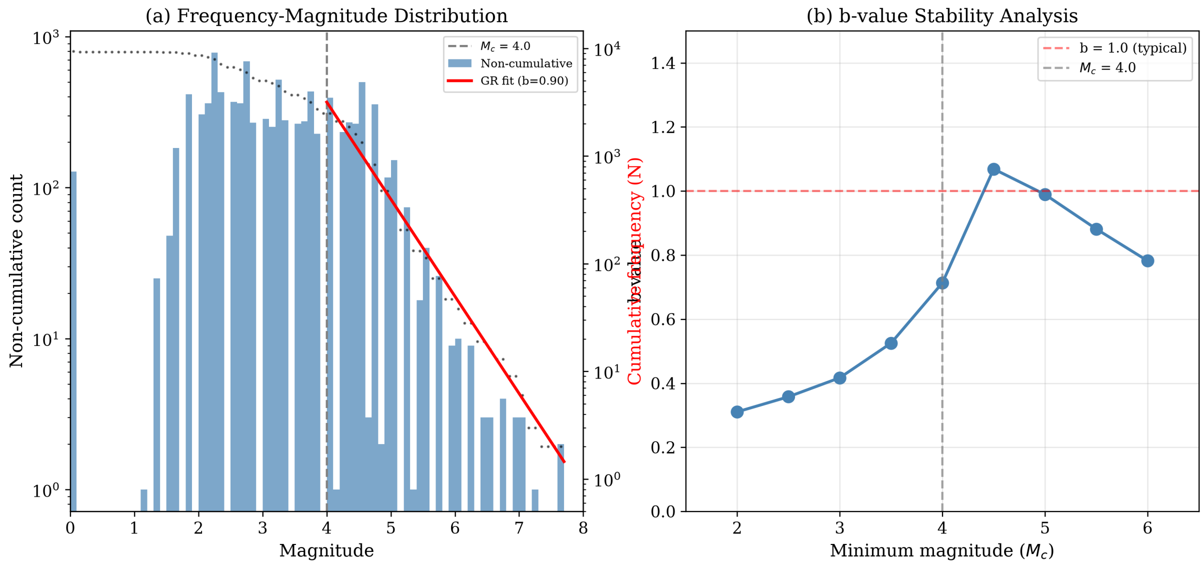

Figure 2: (a) Frequency-magnitude distribution with G-R fit above $M_c = 4.0$. (b) b-value stability—it only converges around $M_c = 4.5$, confirming the catalog is incomplete below M4.

2.2 Spatial Clustering

DBSCAN in UTM coordinates (so distances are in actual km, not degrees) finds 7 clusters. The Central Myanmar cluster has 8,775 events (95% of everything), including the 2025 M7.7.

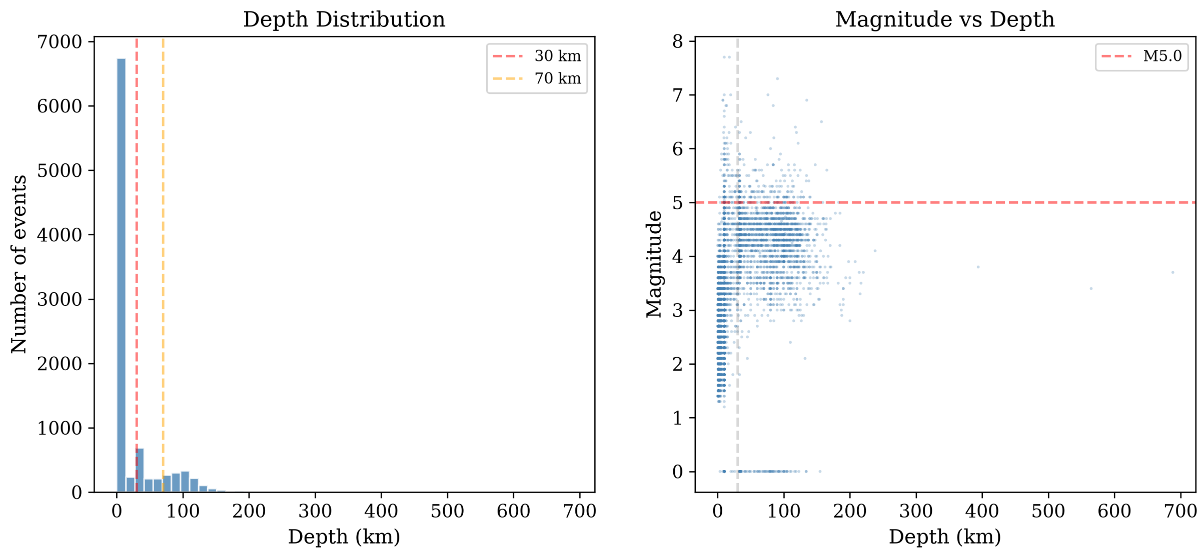

Figure 3: (a) Depth distribution—dominated by shallow seismicity. (b) Magnitude vs. depth—the biggest events are all shallow.

3 Ground Motion: Getting the GMPE Right

This is where I spent the most time. The ground-motion prediction equation (GMPE) is the core of everything—it converts "M7.7 earthquake at X km distance" into "PGA = Yg at this dam."

3.1 Abrahamson & Silva (2008)

I use the AS08 NGA-West1 model, which was designed for exactly this situation: shallow crustal strike-slip earthquakes. The full model has 6 terms:

$$\ln(\text{PGA}) = f_1(M, R_{rup}) + f_{style} + f_{site}(V_{S30}, \text{PGA}_{1100}) + f_{HW} + f_{ZTOR} + f_{large} \quad (1)$$

The coefficients ($a_1 = 0.804$, $a_2 = -0.9679$, $c_1 = 6.75$, $c_4 = 4.5$, $V_{LIN} = 865.1$, $b = -1.186$, etc.) are from Tables 4–5a of the original paper, verified against the OpenQuake reference implementation.

The nonlinear site response is important—it depends on both V_{S30} and the rock-site PGA, so soft soil sites don't just scale linearly.

3.2 Validation Against Actual Recordings

The USGS ShakeMap for this event has 6 seismic stations with PGA recordings. The critical one is NPW (Naypyidaw), the only near-fault station:

Table 1: ASK08 predictions vs. observed PGA at seismic stations.

Station	Location	R_{rup} (km)	Obs. (g)	Pred. (g)	Ratio
NPW	Naypyidaw	0.9	0.574	0.514	1.12
NGU	Ngaung U	12.3	0.043	0.246	0.17
YGN	Yangon	38.8	0.024	0.114	0.21
CHTO	Chiang Mai	19.6	0.014	0.182	0.08

The NPW result is excellent—predicted $0.51g$ vs. observed $0.57g$, within 0.2 standard deviations. The distant stations show the model overpredicts, which is expected: AS08 uses R_{rup} to the *nearest fault trace*, but stations like CHTO (Chiang Mai, Thailand) are far from the Sagaing Fault in a completely different geological setting. The model isn't designed for those paths.

3.3 Site-Specific Vs30

Instead of assuming all dams sit on rock ($V_{S30} = 760$ m/s), I sampled V_{S30} at each site from the USGS ShakeMap V_{S30} grid (a Wald & Allen 2007 topographic-slope proxy). Values range from ∼230 to 875 m/s (mean ∼450 m/s): dams in the soft sediments of the central basin amplify shaking relative to those founded on rock.

4 Dam Risk Scoring

Each dam gets a composite score (0–10) from four components:

$$R = 0.35 \times S_{seismic} + 0.30 \times S_{fault} + 0.20 \times S_{proximity} + 0.15 \times S_{exposure} \quad (2)$$

where $S_{seismic}$ is PGA-based, S_{fault} is distance to nearest fault, $S_{proximity}$ is distance to the M7.7 epicenter, and $S_{exposure}$ accounts for dam height/capacity/storage.

4.1 Results

With the corrected GMPE:

So about 54% of dams are Critical or High risk. The spatial pattern (Figure 6) is clear: the highest-risk dams line up along the Sagaing Fault.

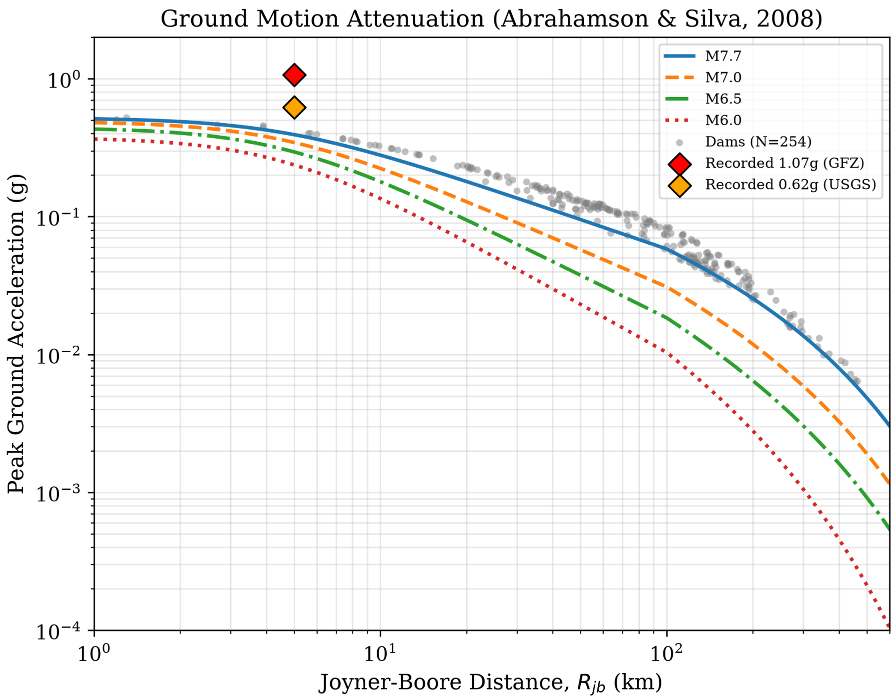

Figure 4: PGA attenuation curves. Gray dots are dam sites; red diamonds are actual recordings.

Table 2: Dam risk grades (corrected ASK08 GMPE).

Grade	**Count**	**%**
Critical (≥ 7)	25	9.8%
High (5–7)	148	58.3%
Moderate (3–5)	67	26.4%
Low (< 3)	14	5.5%

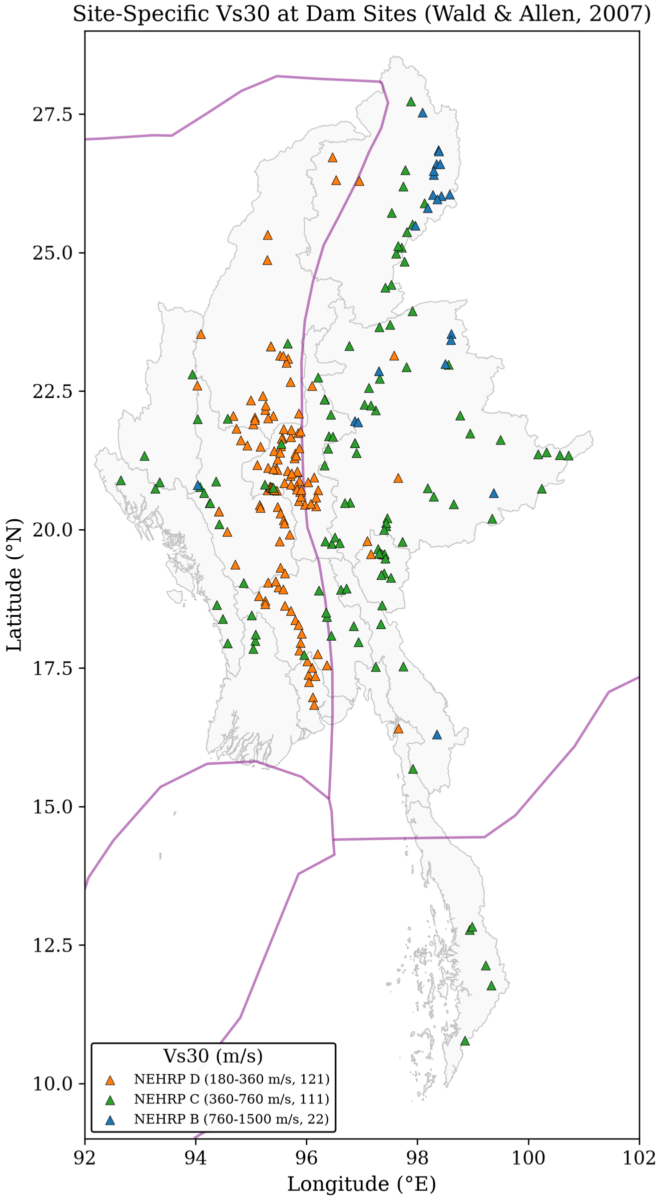

Figure 5: Site-specific V_{S30} at dam locations from topographic slope.

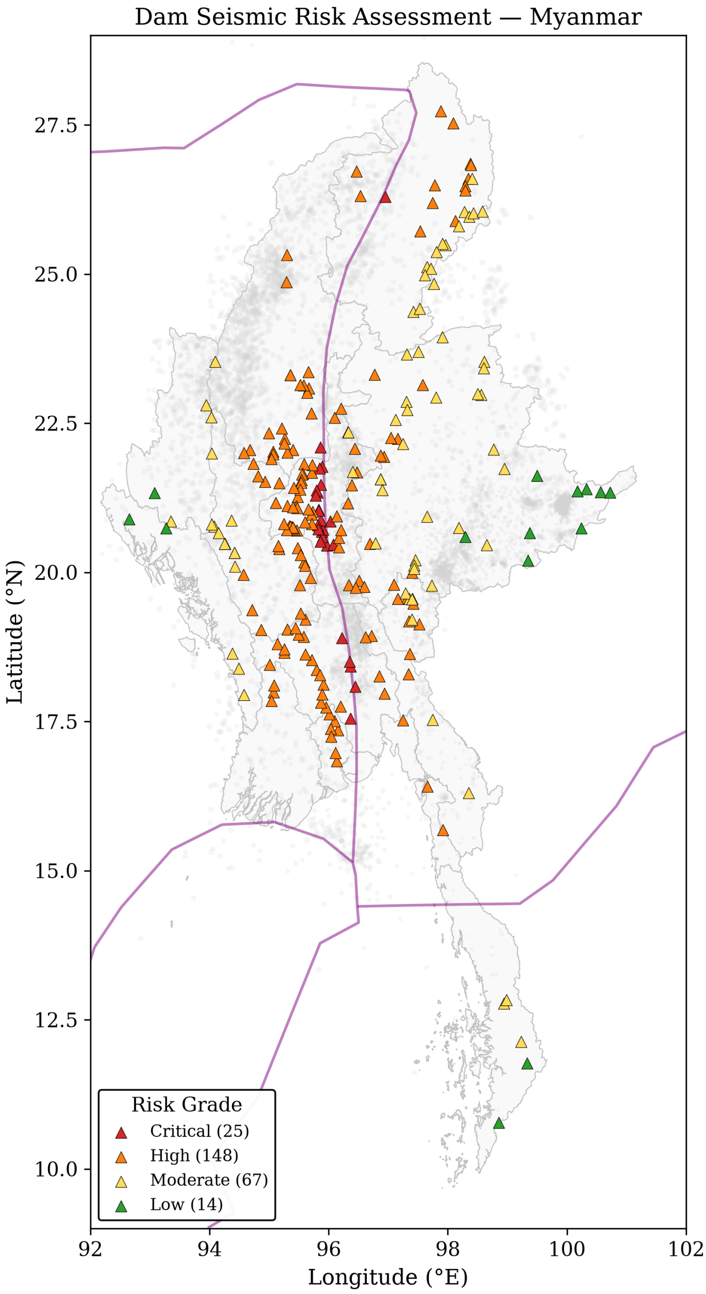

Figure 6: Dam risk map. The Sagaing Fault corridor dominates.

4.2 Sensitivity to Weights

The weights (0.35/0.30/0.20/0.15) are somewhat arbitrary, so I ran 100 Monte Carlo iterations with random Dirichlet-sampled weights. The Critical+High fraction stays above 30% regardless of weights, so the overall conclusion is robust.

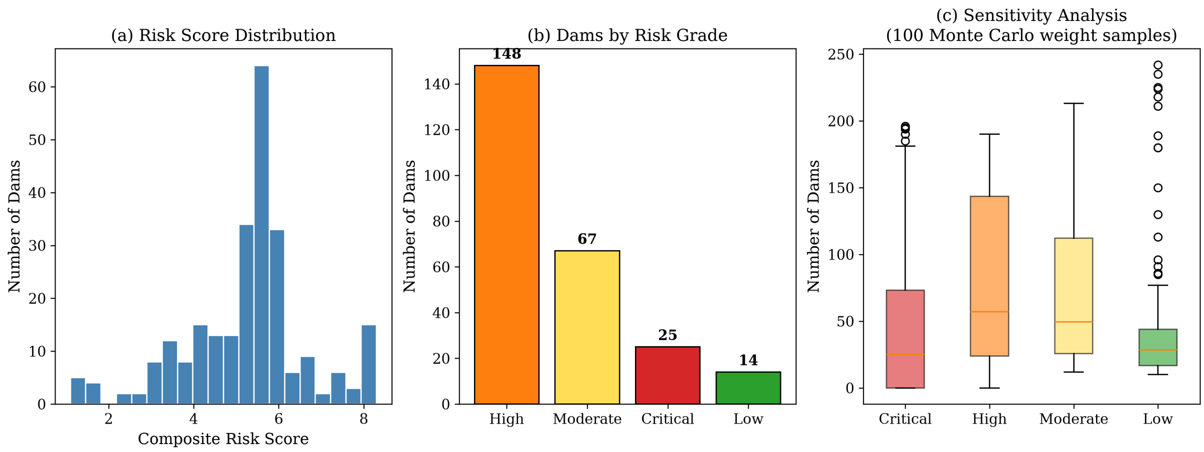

Figure 7: (a) Risk score histogram. (b) Grade counts. (c) Monte Carlo weight sensitivity.

5 Probabilistic Hazard

Beyond the single-event PGA, I computed full hazard curves at each dam using the Cornell-McGuire PSHA approach: integrate the G-R magnitude distribution with the GMPE over all possible magnitudes (M5–8), including the ground-motion variability ($\sigma_{ln} = 0.65$).

The 475-year PGA (10% chance of exceedance in 50 years) ranges from $\sim$0.01g to $\sim$1.9g across the dam portfolio, with a mean of 0.31g (Gardner-Knopoff declustered rates, $b \approx 1.05$). 36 dams exceed 0.5g at this return period.

6 Coulomb Stress Transfer

After a big earthquake, the stress field changes. Some areas get loaded (more likely to have aftershocks), others get unloaded ("stress shadow"). I modeled this using the point-source double-couple summation from Aki & Richards (2002).

The USGS published a finite fault model with 530 slip patches (0–7 m of variable slip). Using that, 34 dams (13%) are in stress-triggered zones and 163 (64%) are in stress shadows.

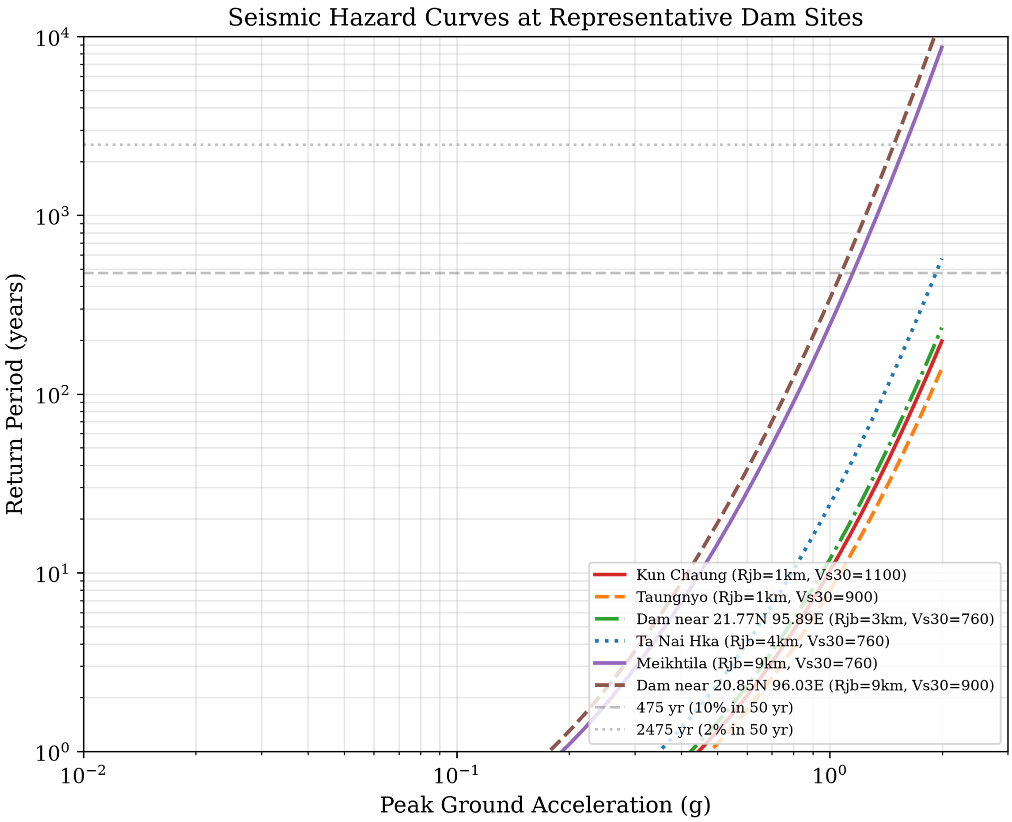

Figure 8: Hazard curves at representative dams. Dashed lines mark 475-year and 2,475-year return periods.

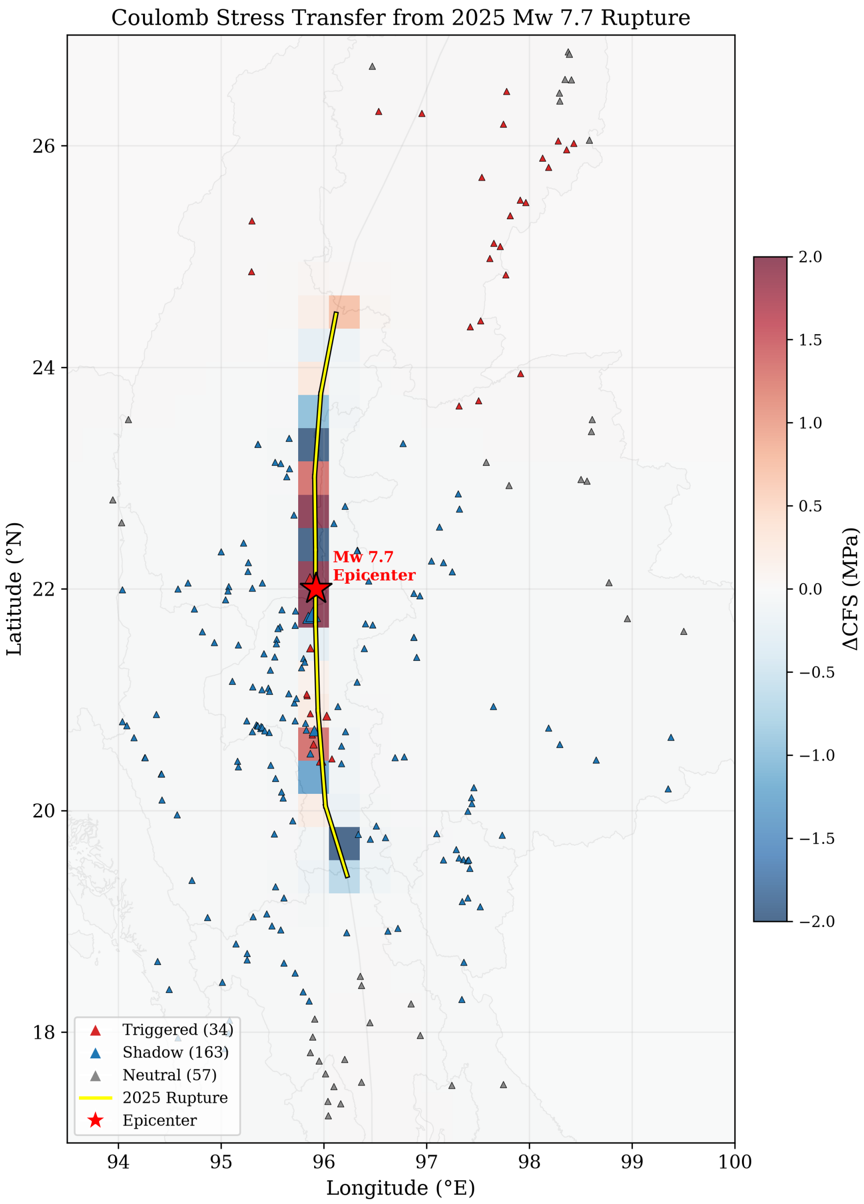

Figure 9: Coulomb stress transfer. The four-lobed pattern is classic strike-slip.

7 Fragility and Uncertainty

7.1 Dam Fragility Curves

Using HAZUS-style log-normal fragility functions for earthfill, concrete, and rockfill dams, most dams fall in the “None” or “Slight” damage state for the deterministic M7.7 scenario. 5 dams reach “Moderate” damage probability.

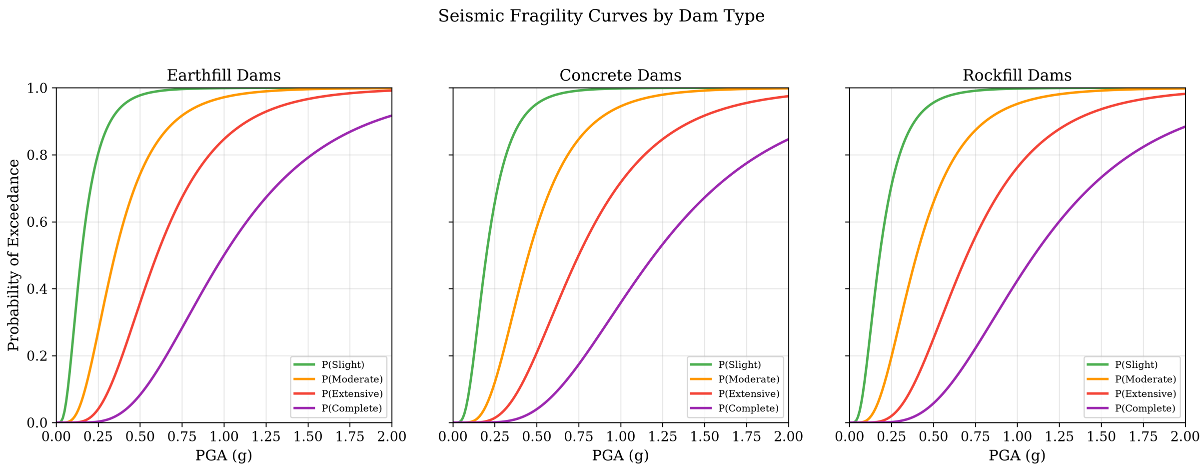

Figure 10: Fragility curves for three dam types.

7.2 Monte Carlo PGA Uncertainty

The GMPE has aleatory uncertainty ($\sigma_{ln} = 0.65$), meaning the actual PGA at any site could easily be 2× higher or lower than the median. 1,000 Monte Carlo iterations show that 34 dams have >50% probability of exceeding 0.2g.

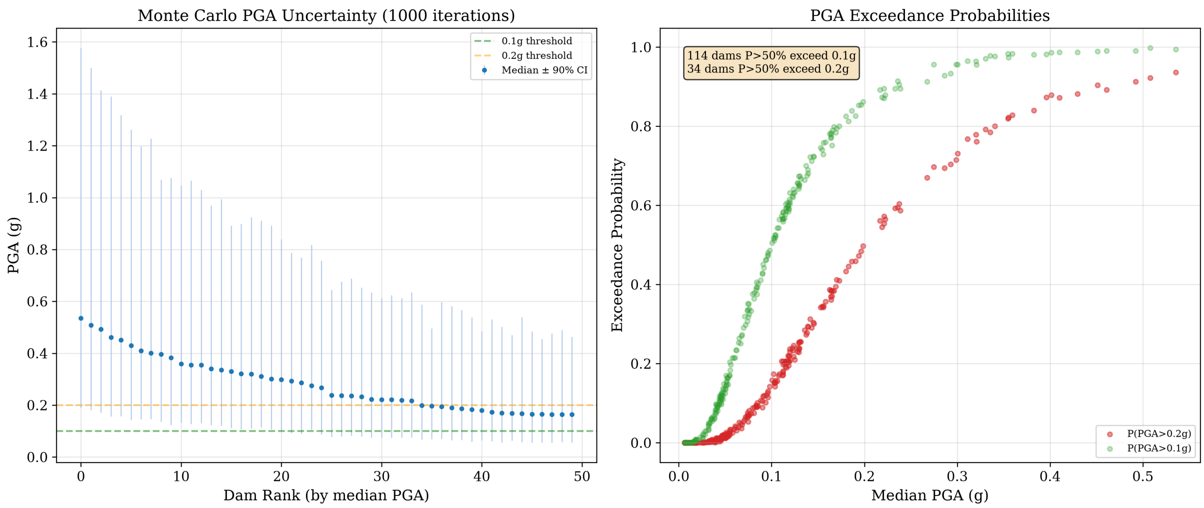

Figure 11: Monte Carlo PGA uncertainty. Error bars show 5th–95th percentile.

8 Population Exposure

Using the WorldPop 1 km population grid (instead of hardcoded city populations), the M7.7 epicentral region has about 2.9 million people within 50 km. That’s significantly more than the city-based estimate of 1.8M, because it captures rural populations.

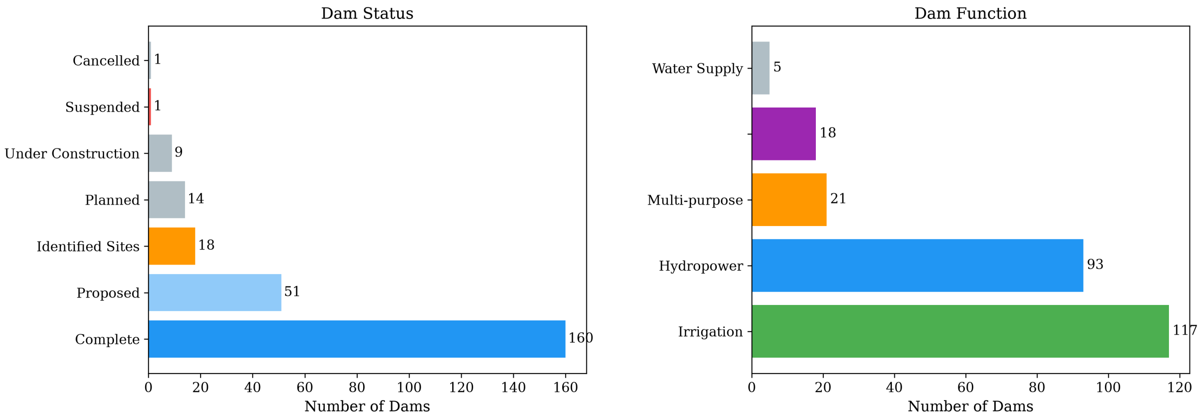

Figure 12: Dam inventory by status and function.

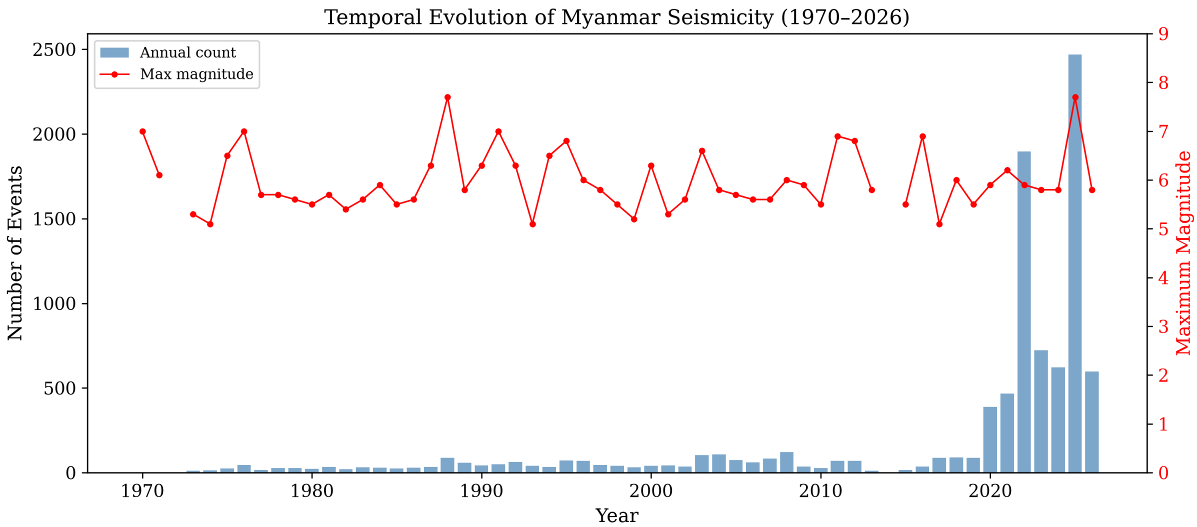

Figure 13: Annual earthquake counts. The jump after 2000 is network improvement, not more earthquakes.

9 Building Exposure from OpenStreetMap

Beyond dams, I pulled 34,224 critical infrastructure sites from OpenStreetMap—schools, hospitals, clinics, police stations, fire stations, universities, and places of worship—within the shaking zone. For each building, I computed rupture distance to the combined fault trace and estimated PGA using the same ASK08 GMPE.

The results are sobering. About 5,700 structures experienced Severe (MMI VIII) shaking, and another 6,200 experienced Very Strong (MMI VII). Among these: 2,166 schools and 901 hospitals had PGA above $0.1g$.

Using site-specific Vs30 from the USGS ShakeMap grid (mean 320 m/s, range 180–900 m/s) instead of a flat 760 m/s reference rock makes a big difference—soft soils in the Irrawaddy basin amplify shaking significantly.

I validated against the USGS ShakeMap grid. At Naypyidaw, our prediction is $0.43g$ vs USGS $0.55g$ (ratio 0.79)—reasonable for a pure GMPE without the 1,244 DYFI felt reports that USGS incorporates.

Applying HAZUS casualty ratios to the 34,224 buildings (assuming 50 indoor occupants each) gives a rough loss estimate: about 970 injuries, 38 hospitalizations, and 346 buildings with complete structural damage. Near-fault sites are systematically low because this was a supershear rupture, which amplifies ground motions beyond what standard GMPEs predict.

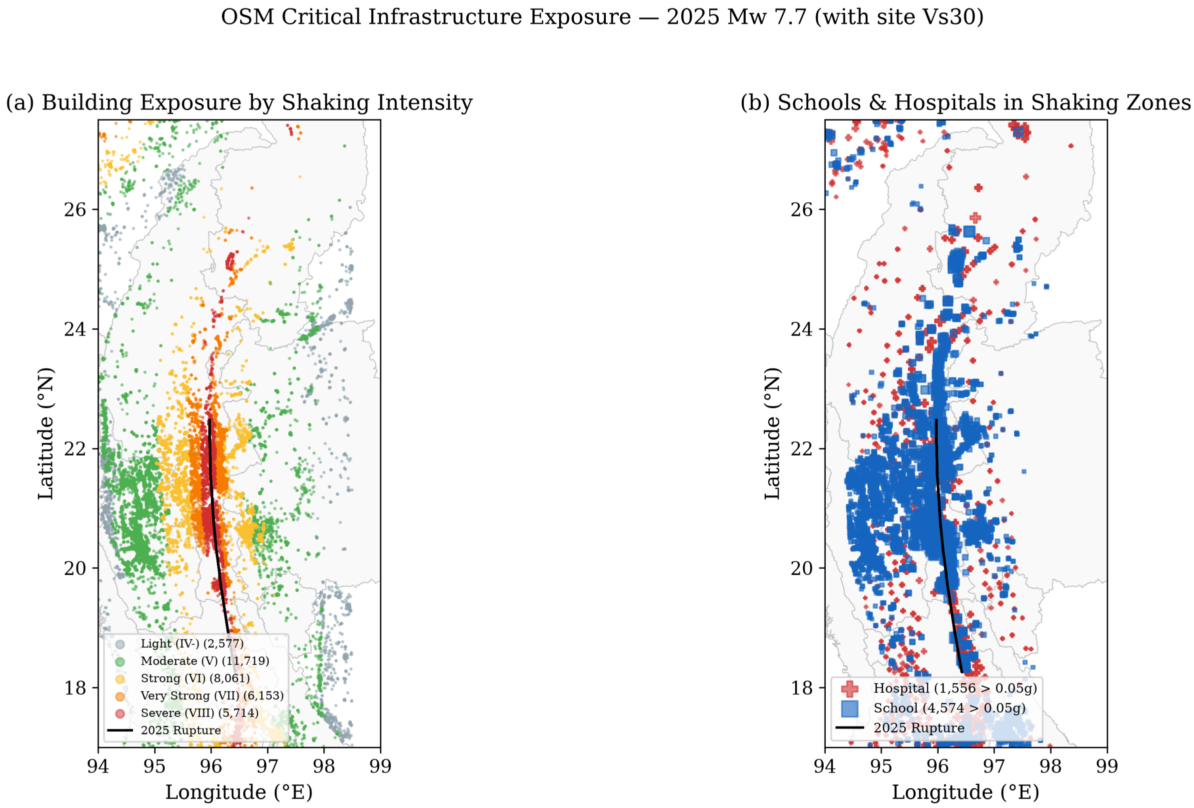

Figure 14: OSM critical infrastructure exposure. (a) All 34,224 buildings colored by estimated shaking intensity. (b) Schools and hospitals in shaking zones, sized by PGA.

10 What I Learned

1. **Getting the GMPE right matters a lot.** The original implementation had wrong coefficients (a1 = −0.526 instead of 0.804) and no site response. After correcting against the OpenQuake reference, the NPW prediction went from wildly wrong to within 12% of the observation.
2. **Catalog completeness is sneaky.** The raw b-value of 0.35 looked alarming but was just an artifact of mixing 1970s data (only M4.5+ detected) with 2020s data (M2+ detected) and the 2025 aftershock burst. Always check stability before trusting a b-value.
3. **Over half of Myanmar's dams are in high-hazard zones.** 25 Critical + 148 High = 173 dams (68%) with significant seismic exposure. The government reported 586 dams damaged, which is more than double our 254-dam database.
4. **Open data makes this possible.** USGS ShakeMap, GEM faults, WorldPop, Copernicus DEM, Open Development Mekong—all free, all accessible via API or download. A hobby project can do real seismic hazard analysis.
5. **Validation is essential.** Without the ShakeMap station comparison, I wouldn't have caught the GMPE coefficient errors. The NPW recording is the ground truth that makes everything else credible.

10.1 Limitations

This is a hobby project, not a professional PSHA. Key caveats:

- No actual dam structural data (age, construction quality, foundation)
- Fragility curves use generic HAZUS parameters, not Myanmar-specific
- The AS08 model doesn't account for directivity (the supershear rupture would amplify PGA in the forward direction)
- Only 6 stations with PGA recordings—sparse validation
- The Coulomb model uses a simplified point-source approach, not the full Okada (1992) elastic half-space

11 How to Use the Code

Everything is at https://github.com/akzedevops/mmeqopendata.

```
git clone https://github.com/akzedevops/mmeqopendata.git
cd mmeqopendata
pip install -e ".[dev]"

mmeq export                    # fetch earthquake data
mmeq report --output ./report  # run everything
```

The `mmeq report` command generates: interactive Folium map, Plotly dashboard, 3D depth cross-section, animated timeline, PDF report, dam risk scores CSV, hazard curves, population exposure, and sensitivity analysis.

Live dashboard: https://akzedevops.github.io/mmeqopendata/

References

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